Sarthak Hazra

Fifth Year BSMS student

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Sarthak Hazra
Homepage

Education

2022–2027 **Bachelor of Sciences and Master of Science (BS-MS)**, *Indian Institute of Science Education and Research, Tirupati*
Department of Mathematics, GPA: 8.5

Academic Interests

Algebraic Geometry

Intersection Theory

Relevant Courses

Analysis	Real Analysis I & II, Complex Analysis, Measure Theory and Integration, Analysis on Euclidean Space, Functional Analysis, ODE
Algebra	Abstract Algebra, Rings & Modules, Fields and Galois Theory, Commutative Algebra, Classical Groups, Representation theory of finite groups
Topology & Geometry	Topology, Algebraic Topology I & II,Differentiable manifold and Lie groups
Other	Elementary Number Theory

Projects and Experience

Dec–Apr 2025 **Semester Reading Project**, *Representation Theory*

- **Key Topics:** Group representations, Maschke's theorem, modules over group algebras, irreducible representations, character theory, Schur's lemma, representation decomposition, orthogonality relations, induced representations, tensor product of representations
- **References:** Linear Representations of Finite Groups by JP Serre, lectures by Prof. DS Nagraj
- **Outcomes:** Developed understanding of finite group representations, character theory, and induced representations, and learned to analyze representations using orthogonality and decomposition principles.

***Advisor**: Prof. D.S Nagaraj*

May–Jul 2025 **Summer Project**, *De-Rham Cohomology and Characteristics Classes*

- **Key Topics:** Exterior algebra, differential forms, chain complexes, De Rham cohomology, Mayer-Vietoris sequences, Homotopy invariance, smooth manifolds, and integration of differential forms.
- **References:** "From Calculus to Cohomology" by Madsen and Tornehave
- **Outcomes:** Ability to translate analytical problems into cohomological formulations

***Advisor**: Dr. Subhash B*

Aug–Dec 2025 **Semester Project**, *Algebraic Geometry*

- **Key Topics:** Algebraic sets, Hilbert's Nullstellensatz, Zariski topology, Regular functions and regular maps, dimension theory, ringed space structure of an Affine algebraic variety, local study of tangent space, singularities, tangent cone, regularity
- **References:** JS Milne's lecture notes

***Advisor**: Dr. Girja Shanker Tripathi*

Jan–Apr 2026 **Semeser Project**, *Introduction to Scheme's*

- **Key Topics:** Sheaves and morphisms of sheaves, construction of the spectrum of a ring $\operatorname{Spec} A$ with the Zariski topology, definitions and properties of schemes, morphisms of schemes, Sheaf of modules, Divisors.
- **References:** Algebraic Geometry (Chapter 2) by Robin Hartshorne.

***Advisor:**Prof. D.S. Nagaraj*

June 2026 – Present **MS-Thesis**, *Degrees of Loci in Projective Constructions and their Degenerations*

- **Objective:**Investigate the algebraic mechanisms underlying degree-drop phenomena and develop a framework for tracking the degrees of loci generated through sequences of projective constructions.
- **References:**. Intersection Theory by William Fulton

***Advisor:**Dr. Chetan Tukaram Balwe*

Talks

Given a talk on the proof of Hilbert's Nullstellensatz.

15th apr 2026 Given a talk on **C**lassification Theorem of Principal G-Bundles.(notes)

Extracurricular Activities

Member of IISER Tirupati Math Club and former member of core team

References

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Indian Institute of Science Education and Research Tirupati
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Languages

Bengali Native proficiency
English Professional working proficiency
Hindi Professional working proficiency